Shh: Secretly Managing Your SSH Keys

Shh is an elegant command-line tool designed for **securely managing SSH keys and secrets** with **AWS Secrets Manager**. It ensures **seamless, automated, and encrypted** storage and retrieval of sensitive credentials, making your DevOps workflow more secure and efficient.

🚀 Features

- 🔐 **Secure SSH Key Storage** – Store and retrieve SSH private keys securely from AWS Secrets Manager.
- ⚡ **Fast & Efficient** – Handles key injection into `ssh-agent` on the fly without writing to disk.
- 🔄 **Seamless Integration** – Works effortlessly with AWS, Ansible, and Terraform.
- 🔍 **Advanced Metadata** – Tracks key details, versions, and automatic rotation schedules.
- 🔁 **Key Rotation** – Monitors key age and suggests rotation timeframes for enhanced security.
- 📎 **Public Key Support** – Upload `.pub` keys alongside private keys for seamless key management.
- 🌍 **Region Flexibility** – Configure AWS regions via CLI arguments or environment variables.

📦 Installation

Prerequisites

- AWS CLI installed and configured with appropriate permissions
- `jq` for JSON processing
- `ssh-agent` running on your system
- `git` for cloning the repository

Quick Installation

The easiest way to install Shh is using our installation script:

```
# Install Shh with one command
curl -fsSL https://raw.githubusercontent.com/jenova-marie/shh/main/shh-install.sh | bash
```

This will:

1. Download and install all Shh components
2. Create symlinks in `/usr/local/bin`
3. Set proper permissions
4. Log all installation activities to `/var/log/shh.log`

Manual Review Before Installation

If you'd like to review the installer before running it (recommended):

```
# Download installation script
curl -fsSL https://raw.githubusercontent.com/jenova-marie/shh/main/shh-install.sh -o shh-install.sh

# Make it executable
chmod +x shh-install.sh

# Run the installer
./shh-install.sh
```

Uninstallation

To remove Shh from your system:

```
# Uninstall directly
curl -fsSL https://raw.githubusercontent.com/jenova-marie/shh/main/shh-install.sh | bash -s uninstall

# Or if you have the script locally
./shh-install.sh uninstall
```

Environment Configuration

You can configure Shh with environment variables:

```
# Regional configuration (in order of precedence)
export SHH_REGION="us-east-2"   # Preferred region
export AWS_REGION="us-east-2"   # Alternative
export AWS_DEFAULT_REGION="us-east-2"  # AWS CLI default

# Secret name configuration
export SHH_SECRETS="my-ssh-keys"  # Name of your AWS Secrets Manager secret
```

🛠️ Usage

🔑 Securely Add an SSH Key to AWS Secrets Manager

The `shh-add` tool stores your SSH keys in AWS Secrets Manager with rich metadata:

```
# Basic usage - adds key to AWS Secrets Manager
shh-add ~/.ssh/mykey_ed25519 [property-name] [region]

# Add both private and public keys
```

```
shh-add ~/.ssh/mykey_ed25519 --pub

# Only upload the specified file, skip public key and fingerprint
shh-add ~/.ssh/mykey_ed25519 --only

# Add key to both AWS Secrets Manager and your local SSH agent
shh-add ~/.ssh/mykey_ed25519 --pub

# Skip adding to SSH agent
shh-add ~/.ssh/mykey_ed25519 --no-ssh-add

# Enable verbose debug output
shh-add ~/.ssh/mykey_ed25519 --debug
```

Key Metadata Features

Each key stored by `shh-add` automatically includes metadata with:

- Key type (ed25519, RSA, etc.) and size (bits)
- Creation and update timestamps
- Version tracking for key rotation history
- SSH key fingerprint for agent identification
- Recommended rotation date (90 days from upload)
- Comments from the original key

🔒 Use SSH Keys with `shh`

The `shh` command retrieves keys from AWS Secrets Manager and uses them with SSH:

```
# Basic syntax (uses username_ed25519 key by default)
shh user@hostname [region] [options]

# Specify a key with -i flag (SSH-style)
shh -i mykey_ed25519 user@hostname [region]

# Standard SSH options are passed through
shh -i mykey_ed25519 -p 2222 user@hostname

# Enable debug output
shh --debug user@hostname
```

The `shh` command performs the following steps:

1. Determines which key to use:
    - If specified with `-i`, uses that key name
    - Otherwise, defaults to `username_ed25519` based on the user part of user@host
2. Securely retrieves the key from AWS Secrets Manager
3. Identifies key fingerprint from metadata

4. Checks if the key is already loaded in `ssh-agent`
5. Adds the key to `ssh-agent` in memory (no disk writes) if needed
6. Connects to the specified server

🔍 Manage Keys with `shh-admin`

The `shh-admin` tool helps you manage your secrets in AWS Secrets Manager:

```
# Interactive mode
shh-admin

# List all keys in the specified secret
shh-admin --list
shh-admin -l

# Specify AWS region
shh-admin --region us-east-2
shh-admin -r us-east-2

# Specify secret name
shh-admin --secret my-ssh-keys
shh-admin -s my-ssh-keys

# Show detailed information about a specific key
shh-admin --detail mykey_ed25519
shh-admin -d mykey_ed25519

# Enable debug output
shh-admin --debug
```

Key Management Features

- Displays key metadata including type, size, and rotation schedule
- Shows key version history and update timestamps
- Views public key contents when available
- Creates new AWS Secrets Manager secrets if they don't exist
- Verifies appropriate AWS IAM permissions

🔄 Key Rotation Best Practices

Shh includes key rotation features:

- Each key automatically has a recommended rotation date (90 days after creation)
- The `shh-admin --list` command shows when each key is due for rotation
- When a key is updated, the rotation history is preserved
- Version numbers are incremented automatically on each update

To rotate a key:

1. Generate a new SSH key: `ssh-keygen -t ed25519 -f ~/.ssh/new_key`
2. Add it to AWS Secrets Manager: `shh-add ~/.ssh/new_key mykey_ed25519 --pub`
3. The previous key data is preserved in the rotation history

🌍 AWS Region Configuration

Region priority (from highest to lowest):

1. Command-line argument (e.g., `shh-add ~/.ssh/mykey_ed25519 mykey us-east-2`)
2. `SHH_REGION` environment variable
3. `AWS_REGION` environment variable
4. `AWS_DEFAULT_REGION` environment variable
5. Default fallback (us-east-2)

🔧 Debug Options

All three commands support a `--debug` flag for troubleshooting:

```
shh user@host keyname --debug
shh-add ~/.ssh/mykey --debug
shh-admin --debug
```

🛡️ Security Considerations

- No SSH keys are ever written to disk during retrieval
- Keys are securely transmitted from AWS Secrets Manager to SSH agent in memory
- All AWS connections use your authenticated AWS CLI credentials
- Key fingerprints are stored to verify agent-loaded keys without requiring passphrase entry

🌍 Open Source & Contributions

We welcome contributions, improvements, and suggestions for enhancements.

```
git clone https://github.com/jenova-marie/shh.git
```

Pull requests are welcome!

📝 License

Shh is released under the **MIT License**.

🎨 Logo Idea

A minimalist **SSH keyhole with sound waves**, representing **secrets & security** in a silent yet powerful way. 🔒 🎵

Let me know if you want tweaks or enhancements, babe! 😘 🔥 🚀

AWS IAM Permissions Required

The following AWS IAM permissions are required for Shh to function properly:

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "secretsmanager:CreateSecret",
                "secretsmanager:GetSecretValue",
                "secretsmanager:PutSecretValue",
                "secretsmanager:UpdateSecret",
                "secretsmanager:DescribeSecret",
                "secretsmanager:ListSecrets"
            ],
            "Resource": "arn:aws:secretsmanager:*:*:secret:YOUR-SECRET-NAME-*"
        }
    ]
}
```

Replace `YOUR-SECRET-NAME` with your actual secret name (e.g., `ssh-keys`). For secrets with path-like structures (e.g., `Test/X/123`), use the full path in the resource name.

You can attach this policy to your IAM user or role through the AWS Management Console or AWS CLI.